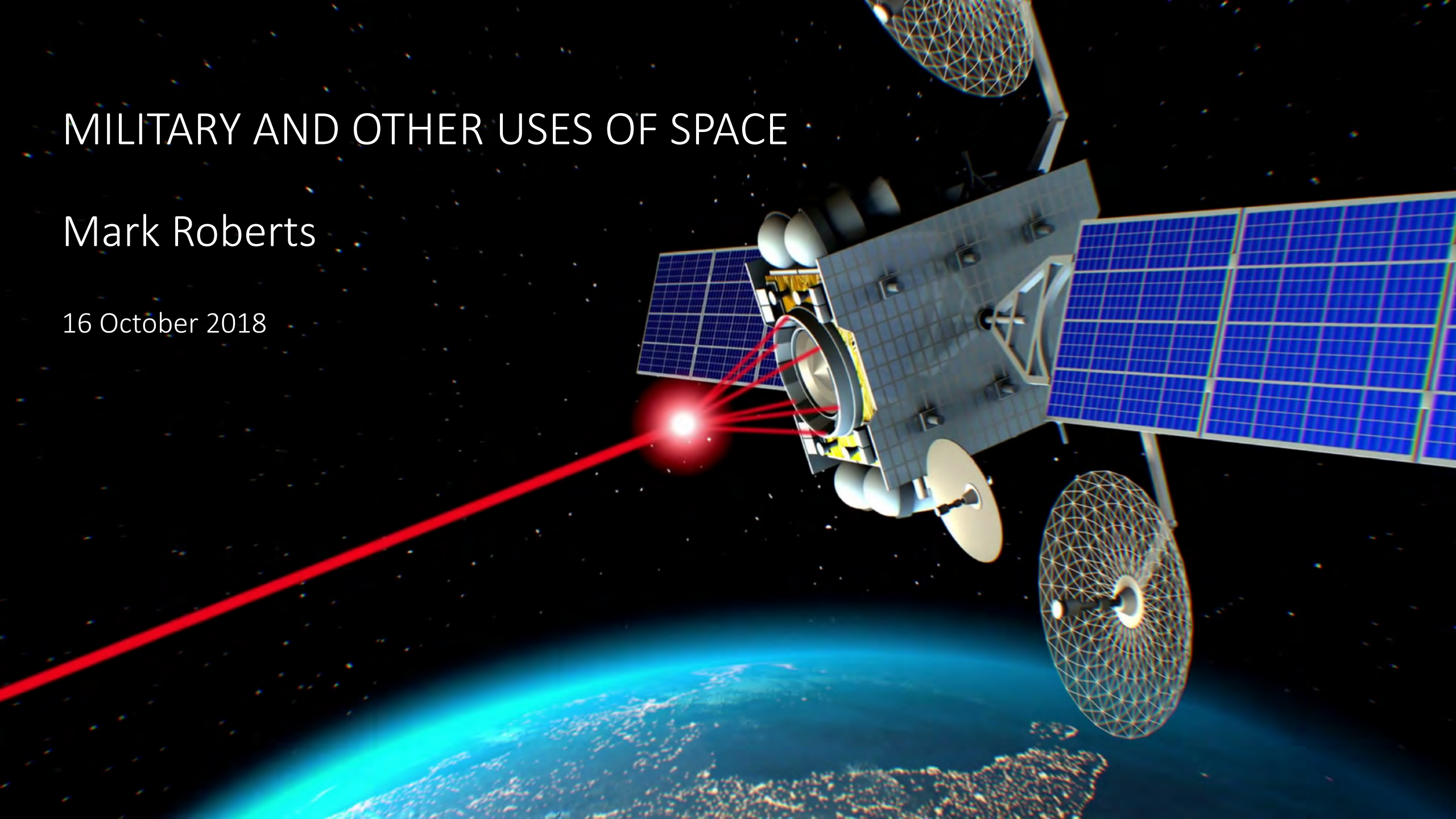
MILITARY AND OTHER USES OF SPACE

Mark Roberts

16 October 2018















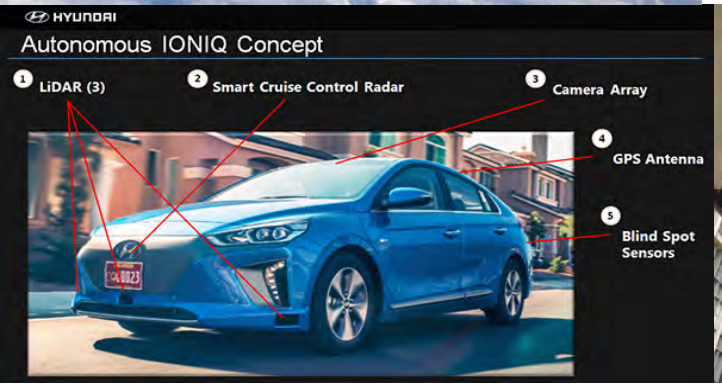






Unmanned “Space” Activity

















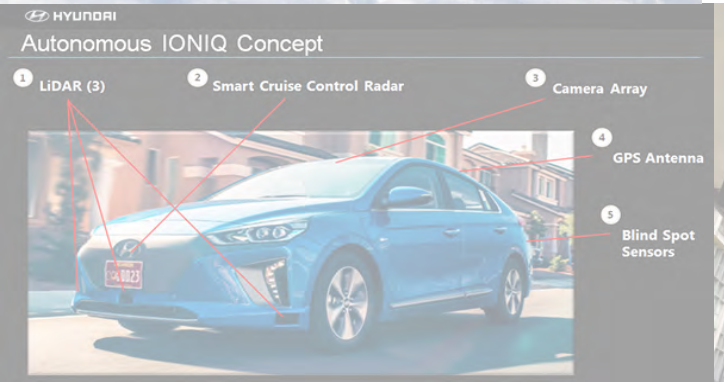




















Unn Activity

Robotics



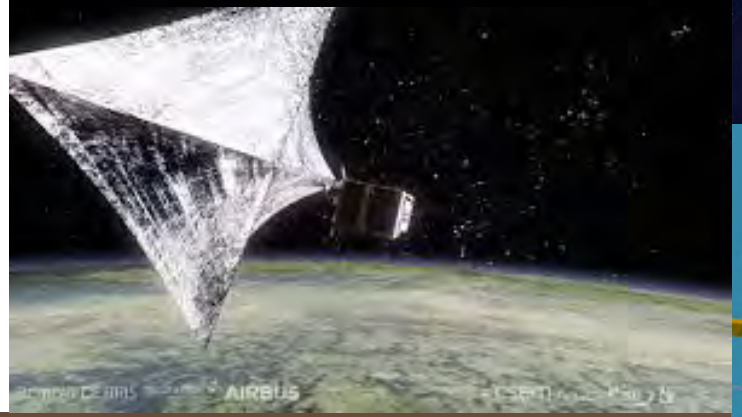
Space Robots

- Capable of surviving environment
- Performing exploration
- Construction, maintenance, servicing
- Designed to be multi-tasking
- *Cheaper* the humans
- Controlled ‘locally’ or ‘remotely’ from earth

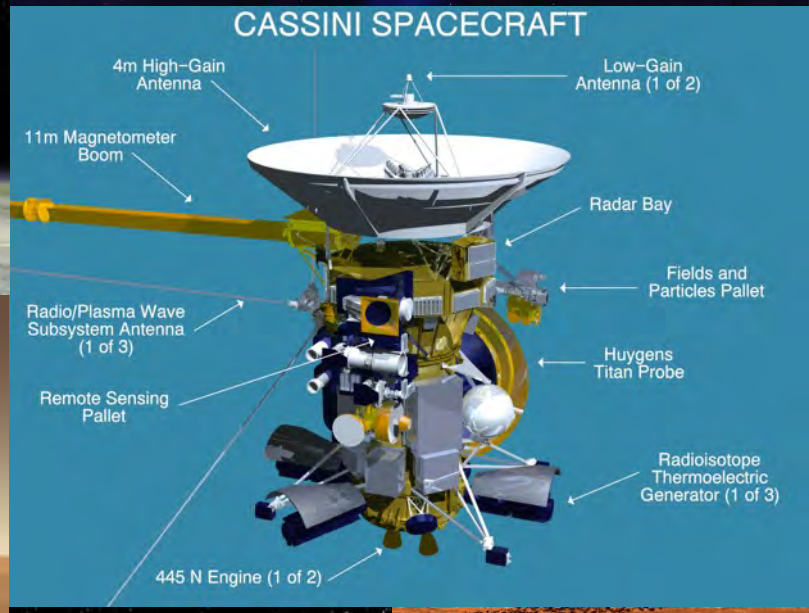
Future (actually now)

- Autonomy and AI
- Navigation/path planning
- Task sequencing
- Task perfection









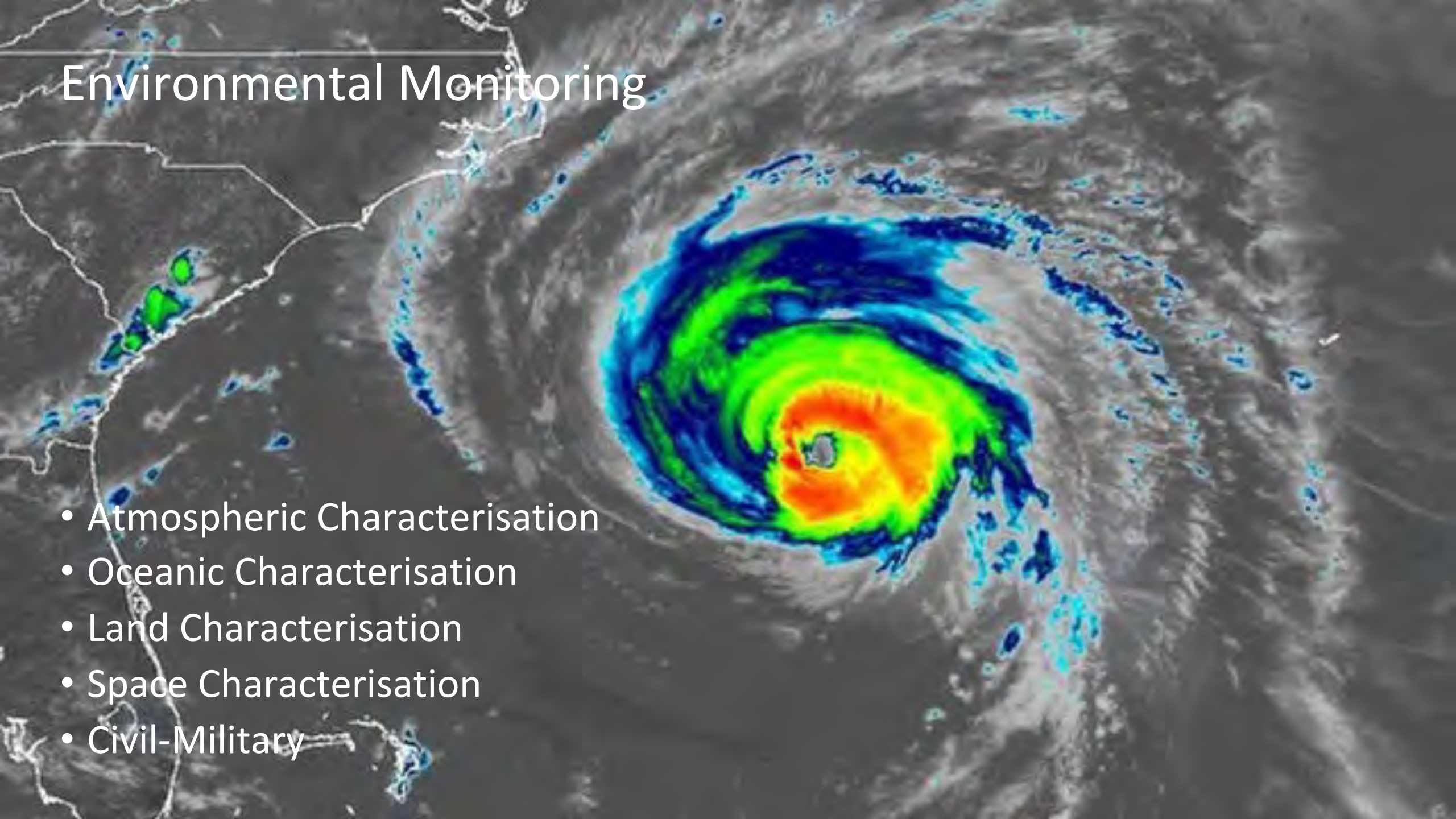








Environmental Monitoring



- Atmospheric Characterisation
- Oceanic Characterisation
- Land Characterisation
- Space Characterisation
- Civil-Military

Atmospheric Characterisation

- Wind – WindSat – Microwave
- Comms/Nav Outage Forecasting

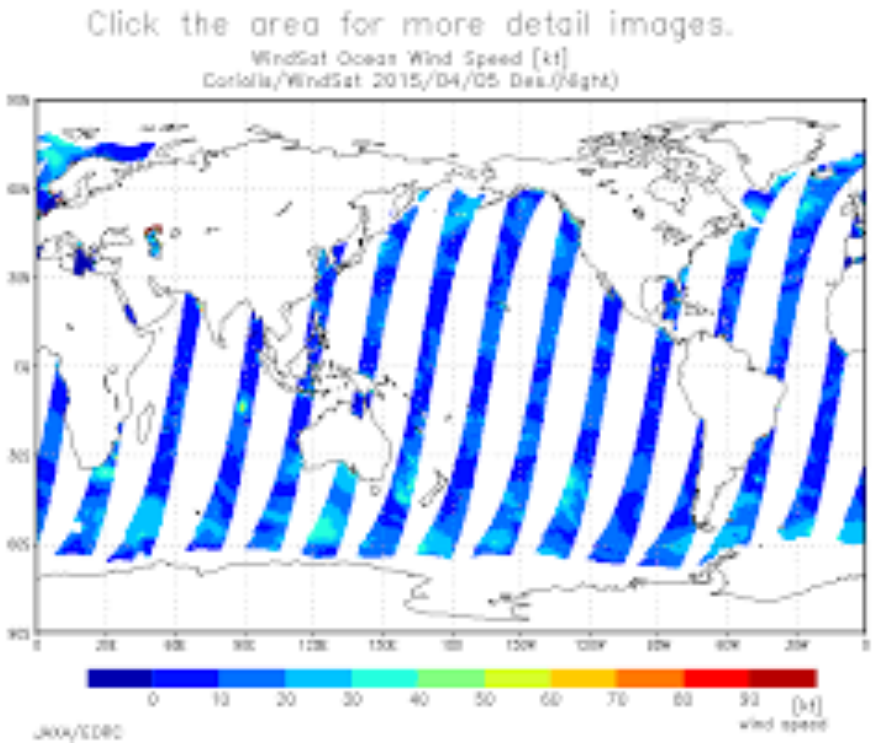
Oceanic Characterisation

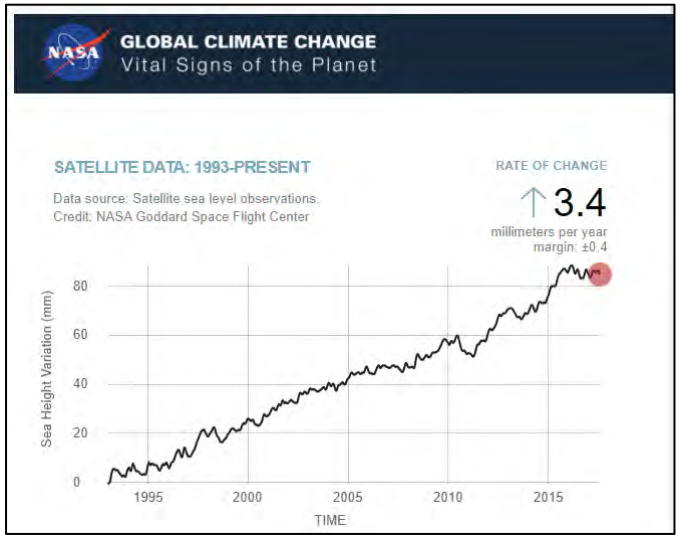
- Sea Levels – Jason 2 - RAlt

Space Characterisation

- Flares / Radiation
- Anomaly resolution





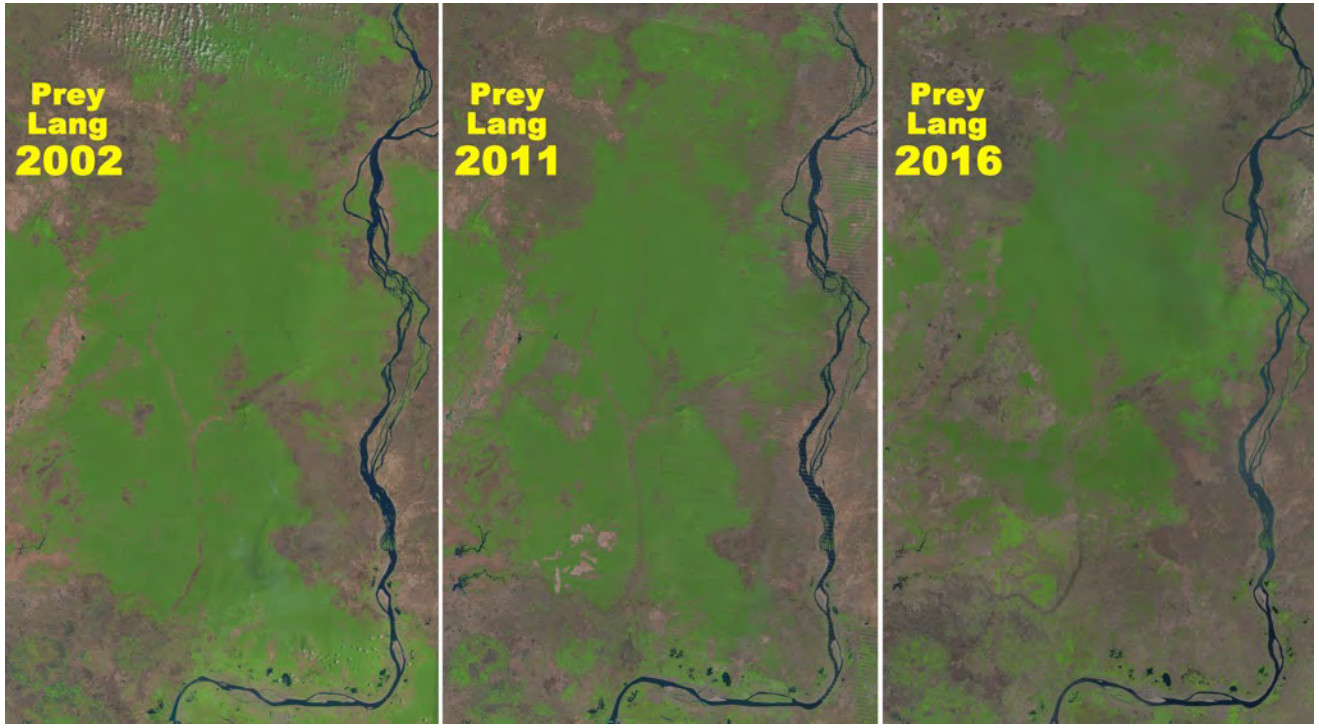


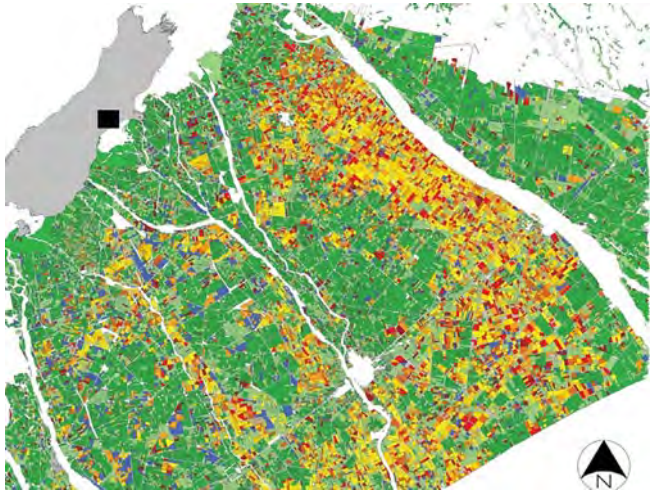


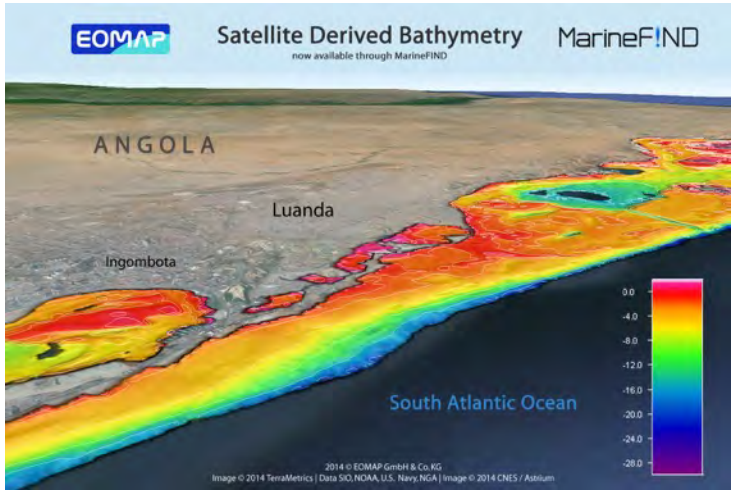


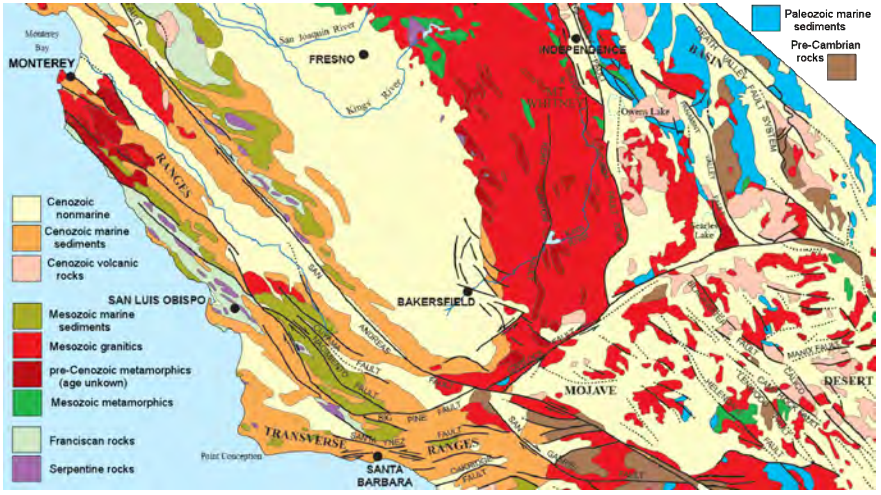
Land Characterisation

- Agriculture / Land use
- Hydrography
- Geology
- Mineralogy
- Land use









Civil-Military

- Forest Fires
- Tsunami Flooding
- Volcanic Activity







Humanitarian

- Remotely sensed data
- Communications
- Logistical planning
- Rapid decision-making
- Resource allocation



Military Use of Space – A Story







Weaponisation of Space



- ASAT Missiles
- On-Orbit Kinetic Actions
- DEW
- Blinding

Is This A Weapon?



Thank You